



# Computational Thinking

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# Solution strategies

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- 1) The “Guess and Check” solution strategy
- 2) The “Go through all the possibilities” solution strategy
- 3) The “Divide the problem into several subproblems” solution strategy
- 4) The “Use of formulas or equations” solution strategy
- 5) The “Discover a structure or pattern” solution strategy
- 6) The “Make a model” solution strategy
- 7) The “Brute force” solution strategy
- 8) The “Divide-and-conquer” solution strategy

# Counting people in a room



# Algorithm for counting people in a room

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```
let N = 0  
for each person in room  
    set N = N + 1
```



# Algorithm for counting people in a room

Testing

```
let N = 0
```

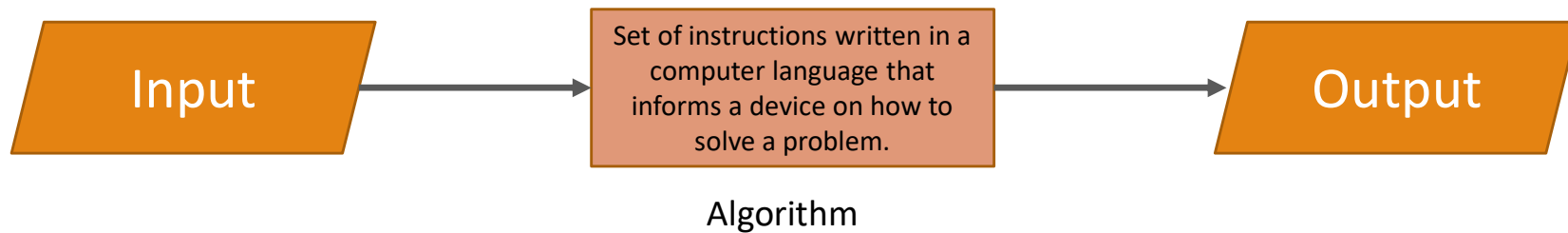
```
for each person in room
```

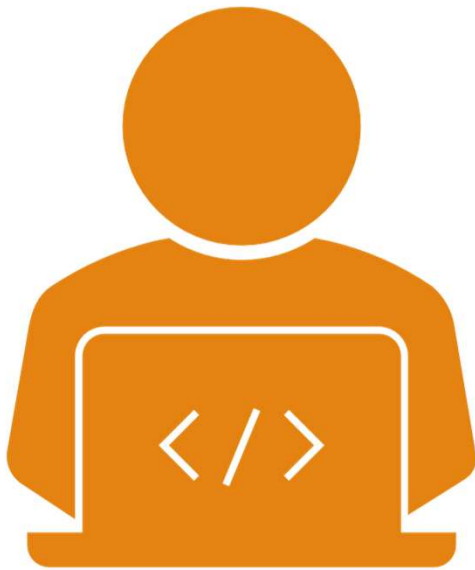
```
  set N = N + 1
```



# What is an algorithm?

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# Formal definition

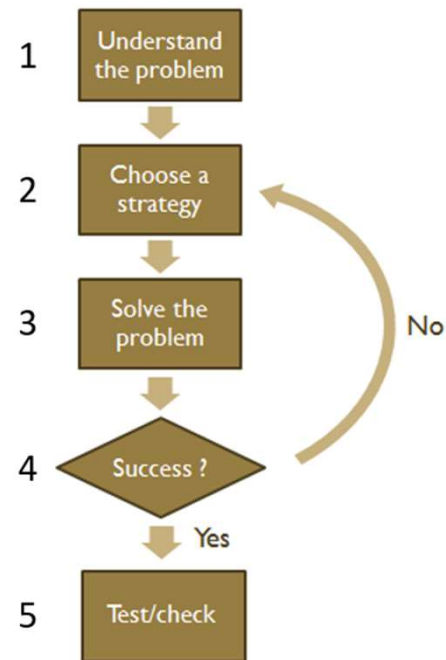
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An algorithm is an effective method to solve a problem or perform a task that consists of a finite number of steps or instructions.

- Algorithms are a special kind of solution strategy
- Algorithms always work

# Algorithm for solving problems

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# Procedure

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- ❑ We will discuss the problem
- ❑ You work together in pairs
- ❑ 5 -10 minutes per problem
- ❑ We discuss the solution

# Problem: Body Mass Index (BMI)

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The BMI is calculated as  $BMI = m/h^2$  where

$m$  = body weight in kilograms and,  
 $h$  = height in meters.

A person with a BMI above  $30 \text{ kg/m}^2$  is considered obese<sup>1</sup>.

Design an algorithm which takes as input a person's height and weight, calculates their BMI and outputs whether the person is obese or not.

Write the pseudocode for this algorithm, then convert it to Python code.

1) <https://www.nhs.uk/common-health-questions/lifestyle/what-is-the-body-mass-index-bmi/>

# Solution: Body Mass Index (BMI) (1/2)

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## STEP-BY-STEP:

- ASK user to input weight in kilograms
- ASK user to input height in meters
- CALCULATE BMI via the formula  
$$\text{BMI} = \text{weight} / \text{height}^2$$
- IF BMI is higher than 30:
  - PRINT “You are in the obese category.”
- ELSE:
  - PRINT “You are not in the obese category.”

## Problem:

The BMI is calculated as  $\text{BMI} = m/h^2$  where

$m$  = body weight in kilograms and,  
 $h$  = height in meters.

A person with a BMI above  $30 \text{ kg/m}^2$  is considered obese.

Design an algorithm which takes as input a person's height and weight, calculates their BMI and outputs whether the person is obese or not.

Draw the flowchart of this algorithm, represent it in pseudocode, then convert it to Python code.

## Solution: Body Mass Index (BMI) (2/2)

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### Code

```
1 weight = float(input("Please give your weight in kilograms. "))
2 height = float(input("Please give your height in meters. "))
3
4 bmi = weight / (height ** 2)
5
6 if bmi > 30:
7     print("You are in the obese category.")
8 else:
9     print("You are not in the obese category.")
10
```